WIKIPEDIA
25 years of the free encyclopedia

Cooking

Cooking, also known as **cookery**, is the art, science and craft of using heat to make food more palatable, digestible, nutritious, or safe. Cooking techniques and ingredients vary widely, from grilling food over an open fire, to using electric stoves, to baking in various types of ovens, to boiling and blanching in water, reflecting local conditions, techniques and traditions. Cooking is an aspect of all human societies and a cultural universal.

A man cooking in a restaurant kitchen, Morocco

Types of cooking also depend on the skill levels and training of the cooks. Cooking is done both by people in their own dwellings and by professional cooks and chefs in restaurants and other food establishments. The term "culinary arts" usually refers to cooking that is primarily focused on the aesthetic beauty of the presentation and taste of the food.

Preparing food with heat or fire is an activity unique to humans. Archeological evidence of cooking fires from at least 300,000 years ago exists, but some estimate that humans started cooking up to 2 million years ago.[1][2]

The expansion of agriculture, commerce, trade, and transportation between civilizations in different regions offered cooks many new ingredients. New inventions and technologies, such as the invention of pottery for holding and boiling of water, expanded cooking techniques. Some modern cooks apply advanced scientific techniques to food preparation to further enhance the flavor of the dish served.[3]

History

Phylogenetic analysis suggests that early hominids may have adopted cooking 1–2 million years ago.[4] Re-analysis of burnt bone fragments and plant ashes from the Wonderwerk Cave in South Africa has provided evidence supporting control of fire by early humans 1 million years ago.[5] In his seminal work *Catching Fire: How Cooking Made Us Human*, Richard Wrangham suggested that evolution of bipedalism and a large cranial capacity meant that early *Homo habilis* regularly cooked food.[6][7] However, unequivocal evidence in the archaeological record for the controlled use of fire begins at 400,000 BCE, much later than the period species like *Homo erectus*

Pots being heated with a wood-burning fire in South India are thought to have lived.[8][9][10] Archaeological evidence from 300,000 years ago,[11] in the form of ancient hearths, earth ovens, burnt animal bones, and flint, are found across Europe and the Middle East. The oldest evidence (via heated fish teeth from a deep cave) of controlled use of fire to cook food by archaic humans was dated to ~780,000 years ago.[12][13] Anthropologists think that widespread cooking fires began about 250,000 years ago when hearths first appeared.[14]

Recently, the earliest hearths have been reported to be at least 790,000 years old.[15]

Communication between the Old World and the New World in the Columbian Exchange influenced the history of cooking. The movement of foods across the Atlantic from the New World, such as potatoes, tomatoes, maize, beans, bell pepper, chili pepper, vanilla, pumpkin, cassava, avocado, peanut, pecan, cashew, pineapple, blueberry, sunflower, chocolate, gourds, green beans, and squash, had a profound effect on Old World cooking. The movement of foods across the Atlantic from the Old World, such as cattle, sheep, pigs, wheat, oats, barley, rice, apples, pears, peas, chickpeas, mustard, and carrots, similarly changed New World cooking.[16]

In the 17th and 18th centuries, food was a classic marker of identity in Europe. In the 19th-century "Age of Nationalism", cuisine became a defining symbol of national identity.[17]

Ilaria Porciani notes that the consequences of industrial food manufacturing, like McDonald's amongst other fast food places, created a desire for an authentic cuisine. She argues that food becomes entangled with nostalgia and comes to be imagined in terms of authenticity and tradition, thus representing continuity with past generations. In this way, food undergoes a process of heritagization. In the 19th and 20th centuries, food became part of a larger process of defining national identity, which can be seen as an informal 'contract' between those who assign heritage status and the people. National intellectualists and folklorists shaped this process by researching national traditions and encouraging them to foster a sense of national unity. Governments, public institutions, cooks, gourmets also participated in this informal contract of building a culinary identity for the nation.[18]

The Industrial Revolution brought mass-production, mass-marketing, and standardization of food. Factories processed, preserved, canned, and packaged a wide variety of foods, and processed cereals quickly became a defining feature of the American breakfast.[19] In the 1920s, freezing methods, cafeterias, and fast food restaurants emerged.

Ingredients

Most ingredients in cooking are derived from living organisms. Vegetables, fruits, grains and nuts as well as herbs and spices come from plants, while meat, eggs, and dairy products come from animals. Mushrooms and the yeast used in baking are kinds of fungi. Cooks also use water and minerals such as salt. Cooks can also use wine or spirits.

Naturally occurring ingredients contain various amounts of molecules called *proteins*, *carbohydrates* and *fats*. They also contain water and minerals. Cooking involves a manipulation of the chemical properties of these molecules.

Carbohydrates

Carbohydrates include the common sugar, sucrose (table sugar), a disaccharide, and such simple sugars as glucose (made by enzymatic splitting of sucrose) and fructose (from fruit), and starches from sources such as cereal flour, rice, arrowroot and potato.[20]

The interaction of heat and carbohydrate is complex. Long-chain sugars such as starch tend to break down into more digestible simpler sugars.[21] If the sugars are heated so that all water of crystallisation is driven off, caramelization starts, with the sugar undergoing thermal decomposition with the formation of carbon, and other breakdown products producing caramel. Similarly, the heating of sugars and proteins causes the Maillard reaction, a basic flavor-enhancing technique.[22]

An emulsion of starch with fat or water can, when gently heated, provide thickening to the dish being cooked. In European cooking, a mixture of butter and flour called a roux is used to thicken liquids to make stews or sauces.[23] In Asian cooking, a similar effect is obtained from a mixture of rice or corn starch and water. These techniques rely on the properties of starches to create simpler mucilaginous saccharides during cooking, which causes the familiar thickening of sauces. This thickening will break down, however, under additional heat.

Fats

Types of fat include vegetable oils, animal products such as butter and lard, as well as fats from grains, including maize and flax oils. Fats are used in a number of ways in cooking and baking. To prepare stir fries, grilled cheese or pancakes, the pan or griddle is often coated with fat or oil. Fats are also used as an ingredient in baked goods such as cookies, cakes and pies. Fats can reach temperatures higher than the boiling point of water, and are often used to conduct high heat to other ingredients, such as in frying, deep frying or sautéing. Fats are used to add flavor to food (e.g., butter or bacon fat), prevent food from sticking to pans and create a desirable texture.

Doughnuts frying in oil

Fats are one of the three main macronutrient groups in human diet, along with carbohydrates and proteins,[24][25] and the main components of common food products like milk, butter, tallow, lard, salt pork, and cooking oils. They are a major and dense source of food energy for many animals and play important structural and metabolic functions, in most living beings, including energy storage, waterproofing, and thermal insulation.[26] The human body can produce the fat it requires from other food ingredients, except for a few essential fatty acids that must be included in the diet. Dietary fats are also the carriers of some flavor and aroma ingredients and vitamins that are not water-soluble.[27]

Proteins

Edible animal material, including muscle, offal, milk, eggs and egg whites, contains substantial amounts of protein.[28][29][30] Almost all vegetable matter (in particular legumes and seeds) also includes proteins, although generally in smaller amounts.[31] Mushrooms have high protein content.[32][33] Any of these may be sources of essential amino acids.[34] When proteins are heated they become denatured (unfolded) and change texture. In many cases, this causes the structure of the material to become softer or more friable – meat becomes *cooked* and is more friable and less flexible. In some cases, proteins can form more rigid structures, such as the coagulation of albumen in egg whites. The formation of a relatively rigid but flexible matrix from egg white provides an important component in baking cakes, and also underpins many desserts based on meringue.

Water

Water is often used to cook foods such as noodles.

Cooking often involves water, and water-based liquids. These can be added in order to immerse the substances being cooked (this is typically done with water, stock or wine). Alternatively, the foods themselves can release water. A favorite method of adding flavor to dishes is to save the liquid for use in other recipes. Liquids are so important to cooking that the name of the cooking method used is often based on how the liquid is combined with the food, as in steaming, simmering, boiling, braising and blanching. Heating liquid in an open container results in rapidly increased evaporation, which concentrates the remaining flavor and ingredients; this is a critical component of both stewing and sauce making.

Vitamins and minerals

Vegetables contain important vitamins and minerals.

Vitamins and minerals are required for normal metabolism; and what the body cannot manufacture itself must come from external sources. Vitamins come from several sources including fresh fruit and vegetables (Vitamin C), carrots, liver (Vitamin A), cereal bran, bread, liver (B vitamins), fish liver oil (Vitamin D) and fresh green vegetables (Vitamin K). Many minerals are also essential in small quantities including iron, calcium, magnesium, sodium chloride and sulfur; and in very small quantities copper, zinc and selenium. The micronutrients, minerals, and vitamins[35] in fruit and vegetables may be destroyed or eluted by cooking. Vitamin C is especially prone to oxidation during cooking and may be completely destroyed by protracted cooking.[36] The bioavailability of some vitamins such as thiamin, vitamin B6, niacin, folate, and carotenoids are increased with cooking by being freed from the food microstructure.[37] Blanching or steaming vegetables is a way of minimizing vitamin and mineral loss in cooking.[38]

Methods

There are many methods of cooking, most of which have been known since antiquity. These include baking, roasting, frying, grilling, barbecuing, smoking, boiling, steaming and braising. A more recent innovation is microwaving[39]. Various methods use differing levels of heat and moisture and vary in cooking time. The method chosen greatly affects the result. Some major hot cooking techniques include: